

◦ RE.6-012 Introduction to Probability - Lecture 7
Textbook Notes - Section 2.6

◦ Section 2.6 - Conditioning (Random Variables)

↳ Ideas are consistent with Event-based Probability.

◦ Conditioning a RV on an Event:

Given an event A with $P(A) > 0$, the conditional PMF of X given A :

$$P_{X|A}(x) = P(X=x|A) = \frac{P(\{X=x\} \cap A)}{P(A)}$$

→ Mechanically - two steps

1. Zero out all probability mass outside A .
2. Renormalise so everything sums to 1

It's a legitimate PMF - the conditioning just restricts your universe to outcomes A .

E.g.: $X = \text{roll of fair dice}$, $A = \text{roll is even}$

$$P_{X|A}(k) = \begin{cases} 1/3 & k = 2, 4, 6 \\ 0 & \text{otherwise.} \end{cases}$$

↳ Three equally likely outcomes remain after restricting to even rolls, each get $1/3$.

◦ Total Probability Theorem

If $A_1, \dots, A_n$ partition the sample space:

$$P_X(x) = \sum_{i=1}^n P(A_i) \cdot P_{X|A_i}(x)$$

↳ Divide & Conquer approach: compute conditional PMF in each scenario, then weight and sum.

• Conditioning on Another RV

When $Y=y$ is observed, the conditional PMF of X given $Y=y$:

$$P_{X,Y}(x|y) = \frac{P_{X,Y}(x, y)}{P_Y(y)}$$

→ Links directly to Joint PMF → Fix a row/column in joint PMF table, then renormalise the slice to sum to 1.

→ Key Relationship: rearranging gives the multiplication rule for joint PMFs:

$$P_{X,Y}(x, y) = P_Y(y) \cdot P_{X,Y}(x|y)$$

↳ sequential approach to building joint PMFs

↳ specify marginal of Y , then conditional distribution of X given each value of Y .

• Conditional Expectation:

The conditional expectation given event A :

$$E[X|A] = \sum_x x \cdot P_{X|A}(x).$$

The conditional expectation given $Y=y$:

$$E[X|Y=y] = \sum_x x \cdot P_{X|Y}(x|y)$$

↳ These are ordinary expectations computed in the restricted universe defined by the conditioning.

→ The Expected Value Rule Extends:

$$\text{↳ } E[g(X)|A] = \sum_x g(x) \cdot P_{X|A}(x)$$

• Total Expectation Theorem :

Most useful result in this section.

↳ Three forms :

$$E[X] = \sum_{i=1}^n P(A_i) E[X|A_i]$$

$$E[X] = \sum_y P_Y(y) \cdot E[X|Y=y]$$

$$E[X|B] = \sum_{i=1}^n P(A_i|B) \cdot E[X|A_i \cap B]$$

↳ Same idea :

→ The conditional average equals the weighted average of conditional averages.

↳ Divide & Conquer for Expectations

↳ Break a hard E into simpler conditional ones

↳ Weight E's by their probability

↳ Sum.

• Geometric Mean & Variance → Payoff Example :

Total expectation theorem is how you derive Geometric distribution's mean + var, without directly computing infinite sums.

$X \rightarrow$ = number of trials until first success, $P(\text{success}) = p$.

→ Condition of first trial :

↳ With probability p : first trial succeeds, $X=1$

↳ With probability $1-p$: first trial fails, problem resets — $X=1+X'$
↳ $X' \rightarrow$ identical fresh copy.

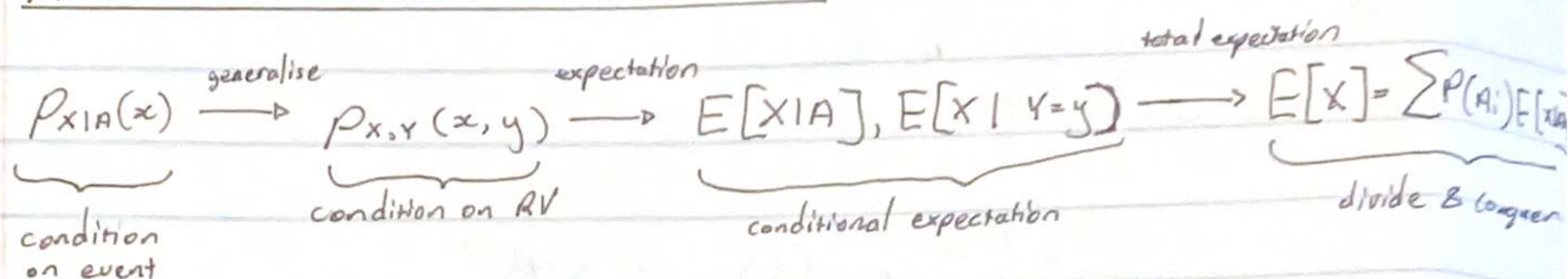
$$E[X] = p \cdot (1) + (1-p)(1 + E[X])$$

$$E[X] = p + (1-p) + (1-p)E[X]$$

$$p \cdot E[X] = 1 \Rightarrow E[X] = \frac{1}{p}$$

$$\text{↳ same trick for: } \text{var}(X) = \frac{1-p}{p^2}$$

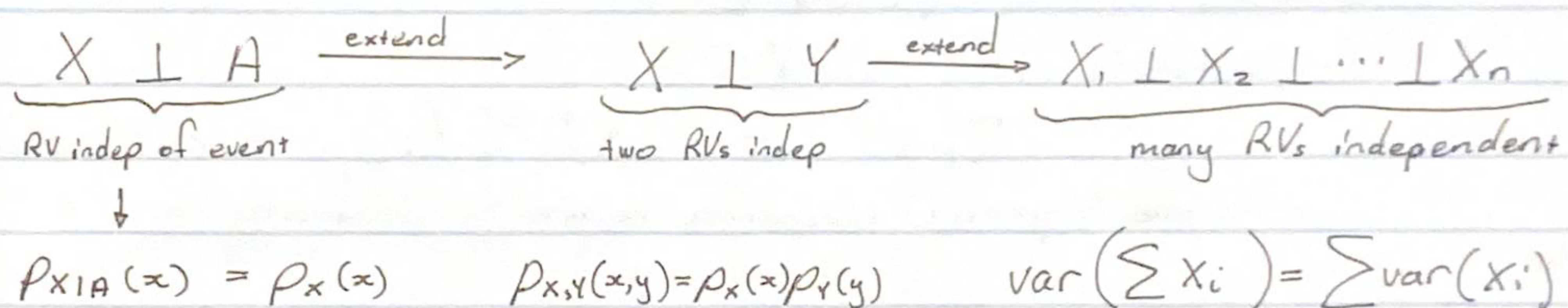
• Architecture of Section 2.6



◦ RES.6-012 Introduction to Probability - Lecture 7
Textbook Notes - Section 2.7

◦ Section 2.7 - Independence (RVs)

• Overview



• Independence of a RV from Event

X is independent of event A if knowing A occurred tells you nothing about X :

$$P_{X,A}(x) = P_X(x) \quad \text{for all } x.$$

↳ Equivalently:

$$P(X=x \text{ and } A) = P_X(x) \cdot P(A)$$

↳ Events $\{X=x\}$ and A are independent for every choice x .

↳ Example:

↳ Two fair coin tosses, X = number of heads, A = number of heads is even

$$\begin{cases} P_X(0) = 1/4, P_X(1) = 1/2, P_X(2) = 1/4 \end{cases}$$

$$\begin{cases} P_{X|A}(0) = 1/2, P_{X|A}(1) = 0, P_{X|A}(2) = 1/2 \end{cases}$$

↳ PMFs differ $\Rightarrow X$ and A are not independent.

↳ Knowing the count eliminates $X=1$.

• Independence of two RVs

X and Y are independent if joint PMF factorises into the product of the marginals

$$P_{X,Y}(x,y) = P_X(x) \cdot P_Y(y) \quad \text{for all } x,y.$$

↳ Equivalents

1. Joint = product of the marginals

2. $\{X=x\}$ and $\{Y=y\}$ are independent events for every (x,y)

3. $P_{X|Y}(x,y) = P_X(x)$ - knowing Y gives no information about X .

→ Conditional independence - X and Y may be dependent overall but independent within a restricted universe defined by event A :

$$P(X=x, Y=y | A) = P(X=x | A) \cdot P(Y=y | A)$$

*

→ NB!!! → Conditional Independence $\neq$ Unconditional Independence

• Benefits of Independence:

→ Expectation of a product:

$$E[XY] = E[X] \cdot E[Y] \text{ if } X, Y \text{ independent.}$$

↳ Not true in general. Dependence requires a covariance term.

→ More generally: For any functions g and h

$$E[g(X) \cdot h(Y)] = E[g(X)] \cdot E[h(Y)]$$

↳ Holds because $g(X)$ and $h(Y)$ are independent.

↳ Functions of independent RVs remain independent.

• Variance of a Sum:

For two independent RVs: $\text{var}(X+Y) = \text{var}(X) + \text{var}(Y)$

For n independent RVs:

$$\text{var}(X_1 + X_2 + \dots + X_n) = \text{var}(X_1) + \text{var}(X_2) + \dots + \text{var}(X_n)$$

• Contrast with Expectation

	Expectation	Variance
Sum	Always $E[X+Y] = E[X] + E[Y]$	Only if indep: $\text{var}(X+Y) = \text{var}(X) + \text{var}(Y)$
Product	Only Only if indep: $E[XY] = E[X] \cdot E[Y]$	No simple Rule

↳ Linearity of Expectation → requires nothing.

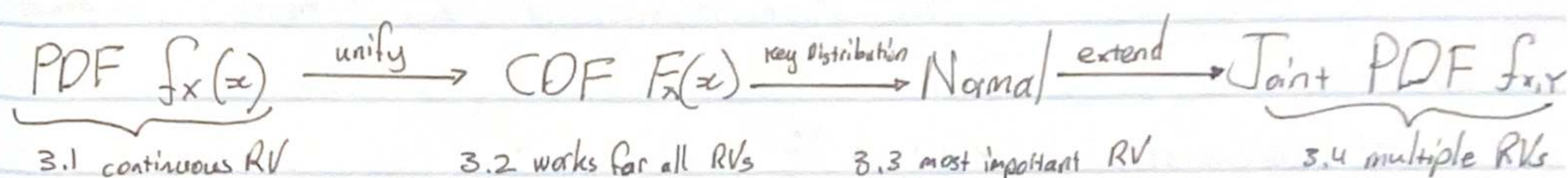
↳ Variance addition → requires independence

Introduction to Probability - Lecture 8

Textbook Notes - Section 3.1 - 3.4

Section 3.1-3.4: General Random Variables

Architecture:



3.1 Continuous Random Variables & PDFs.

For a discrete RV it is possible to list probabilities at specific points.

For a continuous RV, $P(X = a) = 0$ for any single value - the variable takes uncountably many values, so no single point has positive mass.

Instead, described by probability density function (PDF)

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx$$

Probability = area under the PDF curve.

↳ the PDF itself is not a prob → it is prob per unit length at each point x .

↳ Valid PDF requires:

(1) $f_X(x) \geq 0$

(2) $\int_{-\infty}^{\infty} f_X(x) dx = 1.$

↳ continuous versions of PMF requirements

↳ use as a SANITY CHECK

→ Local Interpretation (small δ)

$$P([x, x + \delta]) \approx f_X(x) \cdot \delta$$

↳ $f_X(x)$ → how much probability concentrated near x .

↳ Large value of $f_X(x)$ means outcomes near x are dense.

• Named distributions in 3.1:

→ Uniform on $[a, b]$:

$$f_X(x) = \begin{cases} \frac{1}{b-a} & ; a \leq x \leq b \\ 0 & ; \text{otherwise} \end{cases}$$

↳ Flat density → every subinterval of equal length gets equal prob.
↳ Normalisation constant $\frac{1}{b-a}$ from integral need = 1.

→ Exponential (with λ):

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & ; x > 0 \\ 0 & ; \text{otherwise} \end{cases}$$

$$E[X] = \frac{1}{\lambda} \quad \text{var}(X) = \frac{1}{\lambda^2}$$

↳ Models time until event

↳ Prob of exceed a value decays exponen. $P(X > a) = e^{-\lambda a}$

↳ Note mean = std dev = $\frac{1}{\lambda}$

• 3.1 Expectation & Variance (integrals replace sums)

→ $E[X] = \int_{-\infty}^{\infty} x \cdot f_X(x) dx$

$$\text{var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$$

→ Expected Value Rule:

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) \cdot f_X(x) dx$$

↳ No need to calc $g(x)$ PDF → use $f_X(x)$.

→ Linear Transforms: $Y = aX + b$

$$E[Y] = aE[X] + b \quad \text{var}(Y) = a^2 \cdot \text{var}(X)$$

• 3.2 Cumulative Distribution Functions :

↳ Unify discrete + continuous RVs.

→ Both discrete + continuous RVs have CDF

$$F_X(x) = P(X \leq x)$$

↳ For discrete: $F_X(x) = \sum_{k \leq x} p_X(k)$ → staircase, jumps at each mass

↳ For continuous: $F_X(x) = \int_{-\infty}^{\infty} f_X(t) dt$ → smooth, no jumps.

→ Why CDF

↳ Defined for any RV $\left\{ \begin{array}{l} \rightarrow \text{discrete} \\ \rightarrow \text{continuous} \\ \rightarrow \text{mixed.} \end{array} \right.$

→ CDF Properties

↳ Monotonically nondecrease: $x \leq y \Rightarrow F_X(x) \leq F_X(y)$

↳ $F_X(x) \rightarrow 0$ as $x \rightarrow -\infty$; $F_X(x) \rightarrow 1$ as $x \rightarrow \infty$

↳ Discrete → piecewise constant → staircase

↳ Continuous → smooth → no jumps.

→ 2-way Relationship

Direction	Discrete	Continuous.
CDF → PMF/PDF	$p_X(k) = F_X(k) - F_X(k-1)$	$f_X(x) = \frac{dF_X}{dx}(x)$
PMF/PDF → CDF	$F_X(k) = \sum_{i \leq k} p_X(i)$	$F_X(x) = \int_{-\infty}^x f_X(t) dt.$

→ Maximisation Tool

CDF is a tool for the maximum of multiple RVs

$$X = \max(X_1, \dots, X_n) \Rightarrow F_X(x) = F_{X_1}(x) \cdots F_{X_n}(x)$$

↳ Factorise easily under independence

↳ $1 - F_X(x)$ handles minimums

→ Geometric - Exponential Connection :

↳ CDF shows Exponential (continuous) is the continuous limit of Geometric (discrete)

• 3.3 Normal Random Variables

↳ Most important distribution in Probs + Stats (maybe)

$$f_X(x) = \frac{1}{\sigma \sqrt{2\pi}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

↳ Parameters: μ = mean; σ^2 = variance

↳ Bell curve $\rightarrow$ symmetric μ , spread controlled by σ .

$$E[X] = \mu \quad \text{var}(X) = \sigma^2$$

↳ Why central: sum of many independent, identically distributed RVs $\rightarrow$ builds toward CLT.

$\rightarrow$ Normality is preserved under Linear Transforms

↳ If $X \sim \text{Normal}(\mu, \sigma^2)$ and $Y = aX + b$, then:

$$Y \sim \text{Normal}(a\mu + b, a^2\sigma^2)$$

↳ Normal dist stays Normal $\rightarrow$ with shifted parameters.

$\rightarrow$ Standard Normal Process (and Φ table)

↳ Every Normal reduces to the standard Normal $Y \sim \text{Normal}(0, 1)$

$$\Phi(y) = P(Y \leq y) = \frac{1}{\sqrt{2\pi}} \cdot \int_{-\infty}^y e^{-\frac{t^2}{2}} dt$$

↳ Integral has no closed form solution $\rightarrow$ table.

Step 1 $\rightarrow$ Standardise $\rightarrow Y = \frac{X-\mu}{\sigma}$ is standard Normal

Step 2 $\rightarrow$ Convert $\rightarrow P(X \leq x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$

Step 3 $\rightarrow$ Read the table

→ Key symmetry identity - table only gives $y \geq 0$, recover negatives with
$$\Phi(-y) = 1 - \Phi(y)$$

→ Structure

$$P(a \leq X \leq b) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$$

~~univariate~~

• Joint PDFs of Multiple RVs: 3.4

→ Extending to two continuous RVs

↳ The Joint PDF $f_{X,Y}(x,y)$ satisfies:

$$P((X,Y) \in B) = \iint_B f_{X,Y}(x,y) dx dy$$

↳ Requirements

① $f_{X,Y}(x,y) \geq 0$

② $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx dy = 1$

→ Local interpretation (small δ)

$$P(a \leq X \leq a+\delta, c \leq Y \leq c+\delta) \approx f_{X,Y}(a,c) \cdot \delta^2$$

↳ joint PDF is probability per unit area.

• 3.4 Marginalisation: Recover individual distributions.

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx$$

↳ Integrate away variable you do not want.

→ 2D Uniform Distribution

↳ When (X,Y) is uniform over a region S :

$$f_{X,Y}(x,y) = \frac{1}{\text{area of } S}$$

↳ Probability of any sub-region A is simply $\frac{\text{area of } A}{\text{area of } S}$,
reduces prob problem to a geometry problem.

• 3.4 Joint CDF

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$$

→ PDF recovered by differentiating

$$f_{X,Y}(x, y) = \frac{\partial^2 F_{X,Y}}{\partial x \partial y}(x, y)$$

→ Expected Value Rule

$$E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) \cdot f_{X,Y}(x, y) dx \cdot dy.$$

→ Linearity of Expectation (holds unconditionally)

$$E[aX + bY + c] = aE[X] + bE[Y] + c$$